

DATA PREPROCESSING

Data Preprocessing

- Makes the data more suitable for data mining
- Broad area consisting of a number of different strategies and techniques that are interrelated in complex ways
- Categories:
 - Selecting data objects and attributes for the analysis
 - Creating/changing the attributes

- Aggregation

Sampling

- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

Aggregation

Combining two or more objects into a single object

Purpose

Data reduction

Smaller data sets resulting from data reduction require less memory and processing time, and hence, aggregation may permit the use of more expensive data mining algorithms

Change of scale

By providing high level view of the data instead of a low level

Cities aggregated into regions, states, countries, etc.

Days aggregated into weeks, months, or years

More “stable” Data

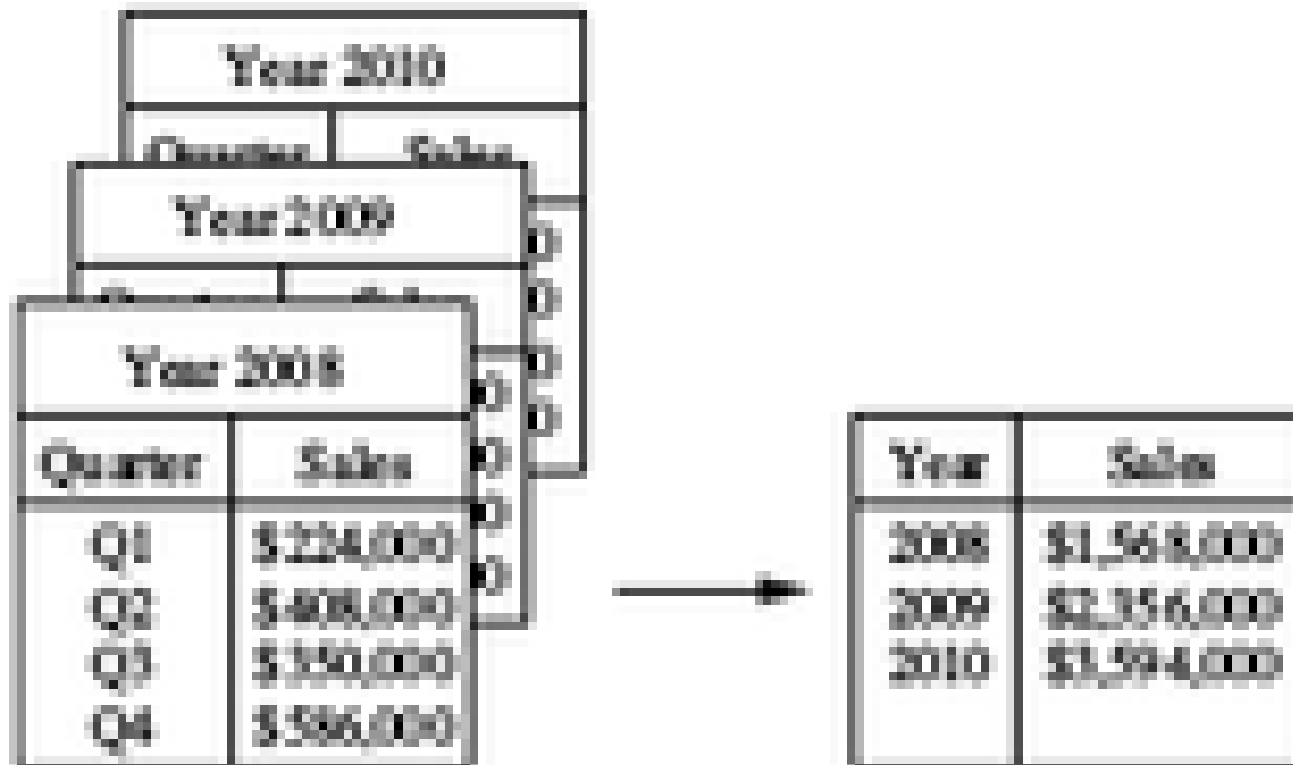
Aggregated data tends to have less variability

Disadvantage:

- Potential loss of interesting details.
- Example: aggregating over months loses information about which day of the week has the highest sales

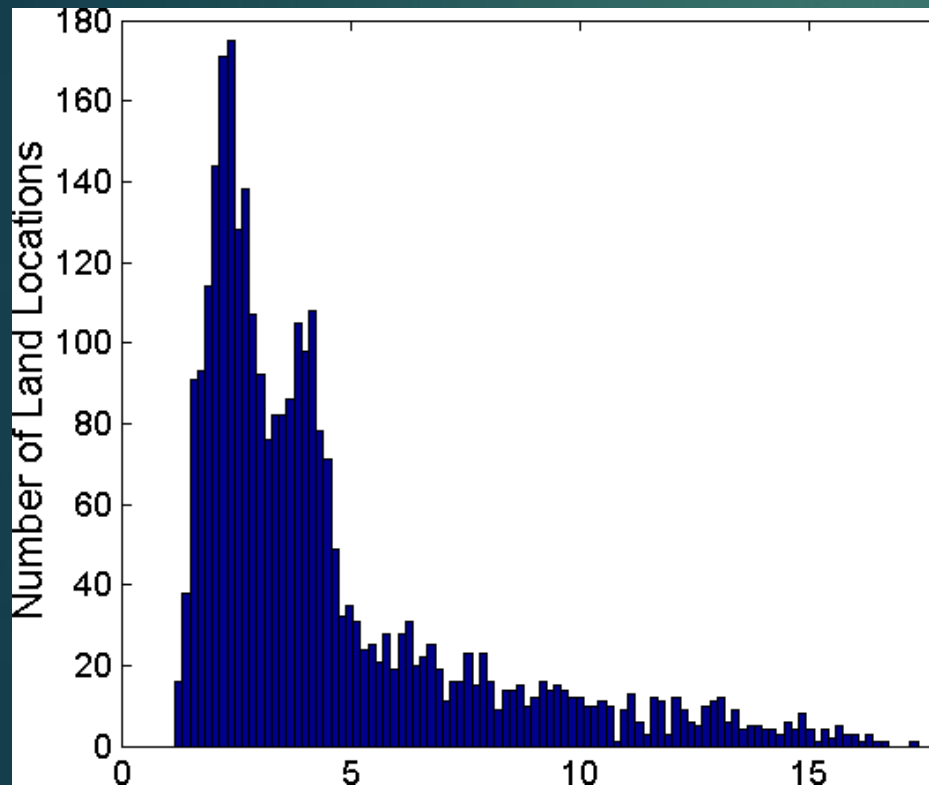


TID	Item	Store Location	Date	Price	...
..	..	..	..	..	..
101	Watch	Delhi	01/10/20	5000	
102	Battey	Mumbai	01/10/20	2000	
103	Shoes	Delhi	02/10/20	2500	

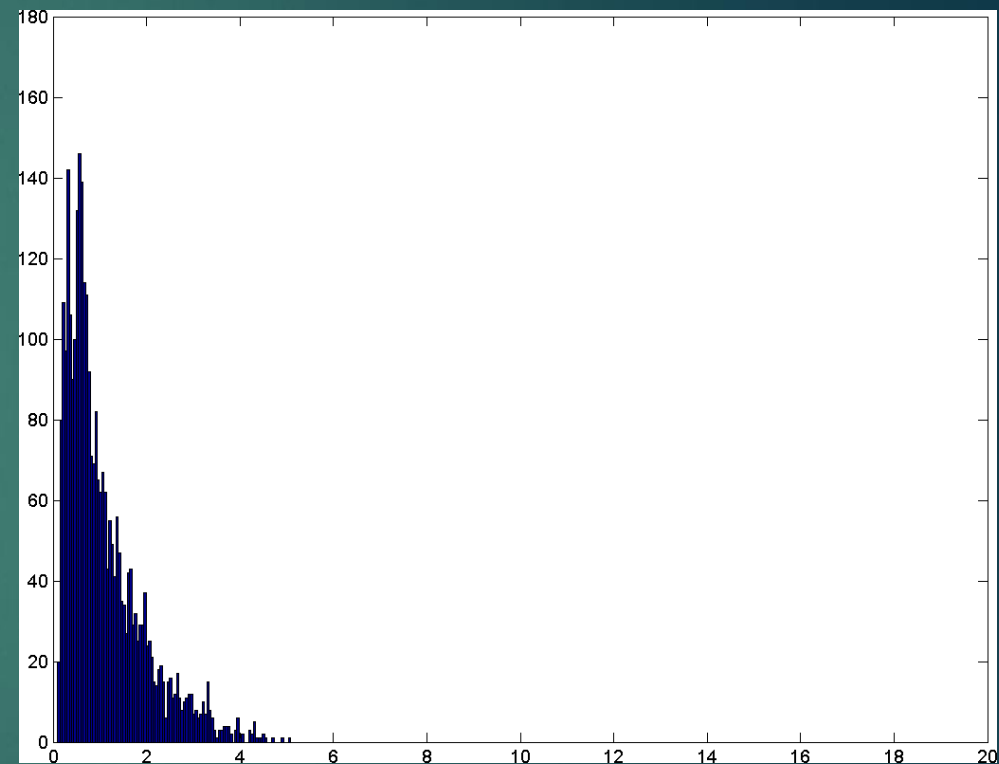


Example: Precipitation in Australia

Variation of Precipitation in Australia



Standard Deviation of Average
Monthly Precipitation



Standard Deviation of Average
Yearly Precipitation

Sampling

- Approach for selecting a subset of the data objects to be analyzed
- Statisticians use sampling because obtaining the entire set of data of interest is too expensive or time consuming
- Data miners sample because it is too expensive or time consuming to process all the data

The key principle for effective sampling :

- Using a sample will work almost as well as using the entire data set, if the sample is **representative**
- A sample is **representative** if it has approximately the same properties (of interest) as the original set of data
- **“Choose a sampling scheme that guarantees a high probability of getting a representative sample”**

Sampling Approaches

Simple Random Sampling

- There is an equal probability of selecting any particular item
- Sampling without replacement
 - As each item is selected, it is removed from the population
- Sampling with replacement
 - Objects are not removed from the population as they are selected for the sample
 - In sampling with replacement, the same object can be picked up more than once

- W h e n the population consists of different types of objects, with widely different numbers of objects method can fail to adequately represent those types of objects that are less frequent
- C a n cause problems when the analysis requires proper representation of all object types
- E x a m p l e : when building classification models for rare classes, it is critical that the rare classes be adequately represented in the sample

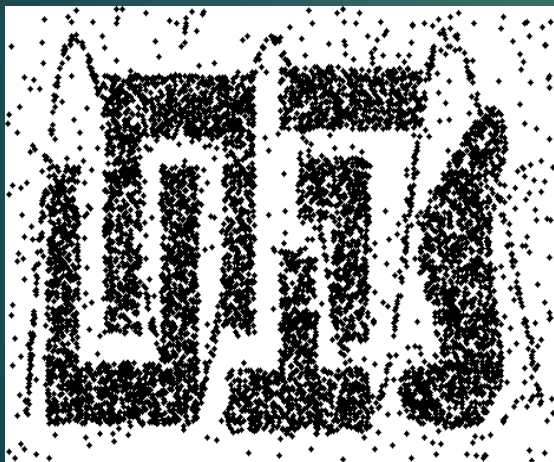
Stratified Sampling

- Split the data into several partitions; then draw random samples from each partition

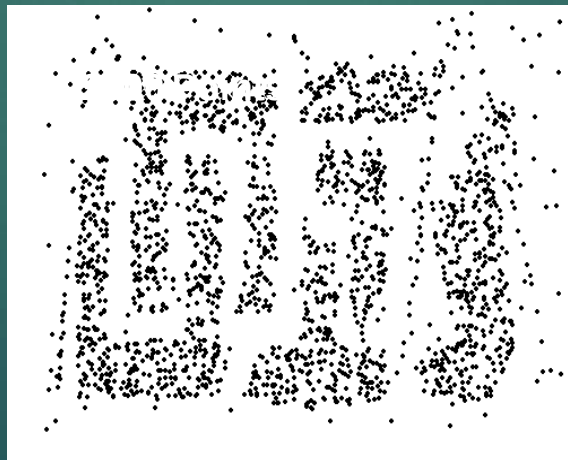
Variations:

- Equal numbers of objects are drawn from each group even though the groups are of different size
- Number of objects drawn from each group is proportional to the size of that group

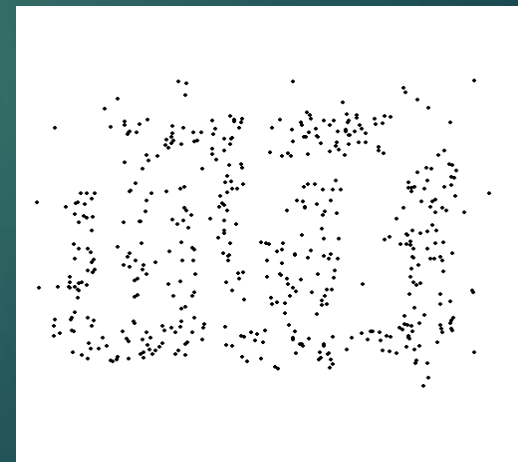
- Once a sampling technique has been selected, it is still necessary to choose the sample size
- Larger sample sizes increase the probability that a sample will be representative, but they also eliminate much of the advantage of sampling
- With smaller sample sizes, patterns may be missed or erroneous patterns can be detected



8000 points



2000 Points



500 Points

Progressive(Adaptive) Sampling

- Proper sample size can be difficult to determine
- **Solution:**
- Start with a small sample, and then increase the sample size until a sample of sufficient size has been obtained
- It requires that there be a way to evaluate the sample to judge if it is large enough

- Although the accuracy of predictive models increases as the sample size increases, at some point the increase in accuracy levels off
- We want to stop increasing the sample size at this leveling-off point
- by keeping track of the change in accuracy of the model as we take progressively larger samples
- by taking other samples close to the size of the current one

Dimensionality Reduction

- Data sets can have a large number of features
- Example: Word count in a document
- **Benefits of Dimensionality reduction**
 - Data mining algorithms work better if the dimensionality is lower
 - We can eliminate irrelevant features and reduce noise
 - Can lead to a more understandable model because the model may involve fewer attributes
 - May allow the data to be more easily visualized
 - Amount of time and memory required by the data mining algorithm is reduced with a reduction in dimensionality

- Techniques that reduce the dimensionality of a data set by creating new attributes that are a combination of the old attributes - Dimensionality reduction

The reduction of dimensionality by selecting new attributes that are a subset of the old - Feature subset selection or Feature selection

Curse of Dimensionality

- Phenomenon that many types of data analysis become significantly harder as the dimensionality of the data increase
- As dimensionality increases, the data becomes increasingly sparse in the space that it occupies

Curse of dimensionality Problems:

- **Classification**
 - There are not enough data objects to allow the creation of a model that reliably assigns a class to all possible objects
- **Clustering**
 - Definitions of density and the distance between points, which are critical for clustering, become less meaningful

Linear Algebra Techniques for Dimensionality Reduction

- **Principal Components Analysis (PCA)**

For continuous attributes that finds new attributes (principal components) that

- are linear combinations of the original attributes
- are orthogonal (perpendicular) to each other
- capture the maximum amount of variation in the data

- **Singular Value Decomposition (SVD)**

Feature Subset Selection

- Reducing dimensionality by using only a subset of features
- Involves removing:
 - **Redundant features** (duplication of much or all of the information contained in one or more other attributes)
 - **Irrelevant features** (contain almost no useful information for the data mining task at hand) are removed
- **Ideal approach:** Try all possible subsets of features as input to the data mining algorithm of interest, and then take the subset that produces the best results
- Number of subsets involving n attributes is 2^n , such an approach is impractical in most situations

Standard approaches to Feature Selection:

Embedded Approach

- Feature selection occurs naturally as part of the data mining algorithm
- During the operation of the data mining algorithm, the algorithm itself decides which attributes to use and which to ignore
- Example: Algorithms for building decision tree classifier

Filter approaches

- Features are selected before the data mining algorithm is run, using some approach that is independent of the data mining task
- Example: selecting sets of attributes whose pairwise correlation is as low as possible

Wrapper approaches

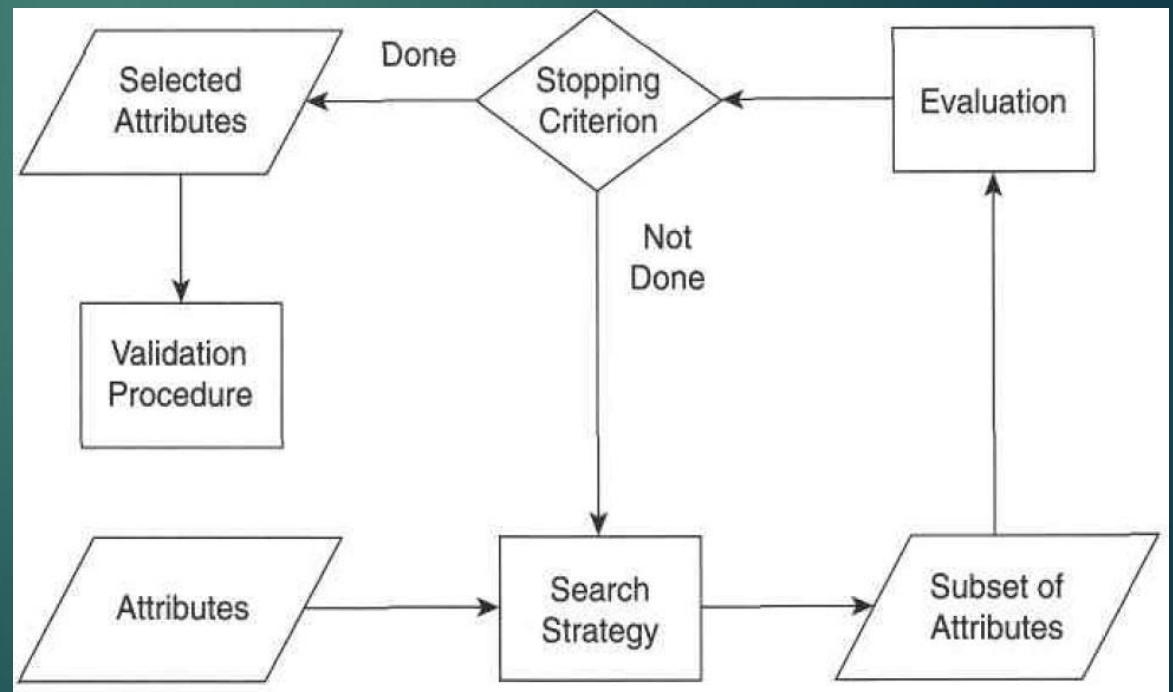
- Use the target data mining algorithm as a black box to find the best subset of attributes

An Architecture for Feature Subset Selection

It is possible to encompass both the filter and wrapper approaches within a common architecture

The Feature Selection process consists of four parts:

- Measure for evaluating a subset
- Search strategy that controls the generation of a new subset of features
- Stopping criterion
- Validation procedure



Search strategy

- Feature subset selection is a search over all possible subsets of features
- Many different types of search strategies can be used
 - should be computationally inexpensive
 - should find optimal or near optimal sets of features
- Usually not possible to satisfy both requirements, and thus, tradeoffs are necessary

Evaluating a subset

- Step to judge how the current subset of features compares to others that have been considered
- Requires an **evaluation measure** that attempts to determine the goodness of a subset of attributes with respect to a particular data mining task, such as classification or clustering
- For a wrapper method, subset evaluation uses the target data mining algorithm
- For a filter approach, the evaluation technique is distinct from the target data mining algorithm

Stopping criterion

- Because the number of subsets can be enormous and it is impractical to examine them all, some sort of stopping criterion is necessary

Based on one or more conditions such as-

- Number of iterations
- Whether the value of the subset evaluation measure is optimal or exceeds a certain threshold
- Whether a subset of a certain size has been obtained
- Whether simultaneous size and evaluation criteria have been achieved
- Whether any improvement can be achieved by the options available to the search strategy

Validation Procedure

- Results of the target data mining algorithm on the selected subset should be validated

Approach 1:

- Run the algorithm with the full set of features and compare the full results to results obtained using the subset of features
- Hopefully, the subset of features will produce results that are better than or almost as good as those produced when using all features

Approach 2:

- Use a number of different feature selection algorithms to obtain subsets of features and then compare the results of running the data mining algorithm on each subset

Feature weighting

- Alternative to keeping or eliminating feature
- More important features are assigned a higher weight, while less important features are given a lower weight
- Weights are sometimes assigned based on domain knowledge about the relative importance of features
- Alternatively, they may be determined automatically
- Example: some classification schemes, such as support vector machines produce classification models in which each feature is given a weight
- Features with larger weights play a more important role in the model

Feature Creation

Creating a new set of attributes from the original attributes, that captures the important information in a data set much more effectively

Methodologies for creating new attributes:

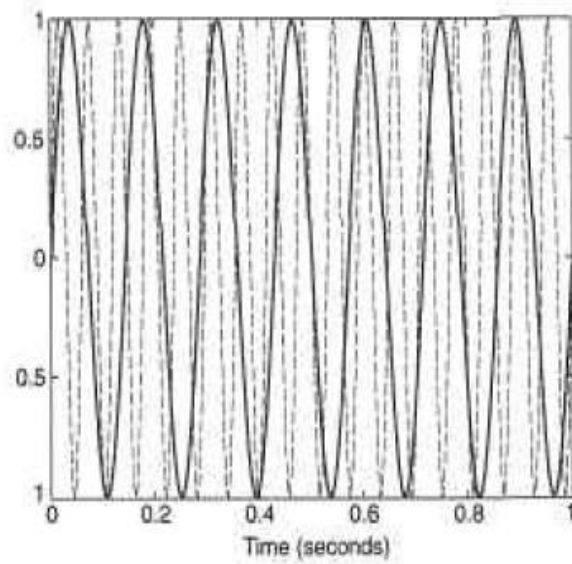
- Feature extraction
- Mapping the data to a new space
- Feature construction

Feature Extraction

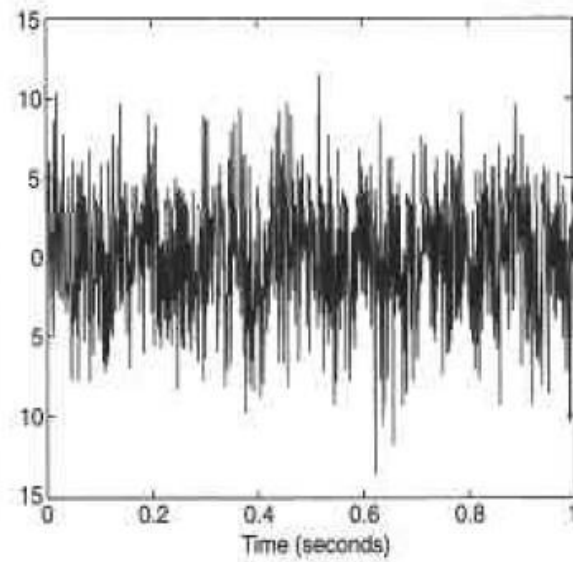
- Creation of a new set of features from the original raw data
- Photographs are classified according to whether or not it contains a human face
- Highly domain-specific
- Whenever data mining is applied to a relatively new area, a key task is the development of new features and feature extraction methods

Mapping the Data to a New Space

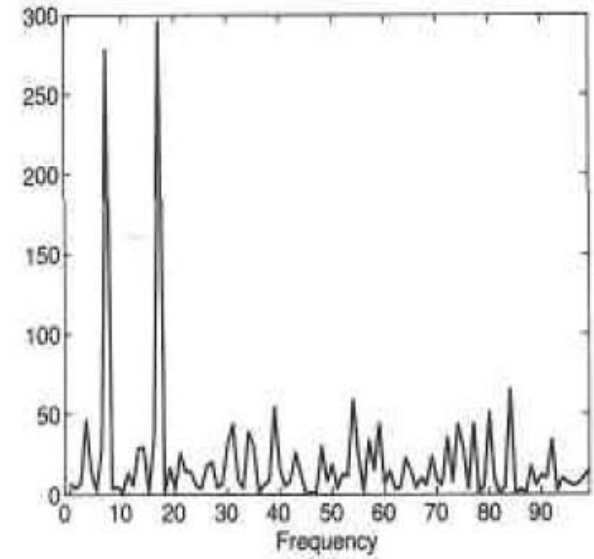
- Totally different view of the data can reveal important and interesting features
- Example: time series data, which often contains periodic patterns
- If there is only a single periodic pattern and not much noise then the pattern is easily detected
- If there are a number of periodic patterns and a significant amount of noise is present, then these patterns are hard to detect
- Such patterns can often be detected by applying a Fourier transform to the time series
- Other transformations: wavelet transformation



(a) Two time series.



(b) Noisy time series.



(c) Power spectrum

Feature Construction

- Sometimes the features in the original data sets have the necessary information, but it is not in a form suitable for the data mining algorithm
- One or more new features constructed out of the original features can be more useful than the original features
- Example: density feature constructed from the mass and volume features

$$\text{density} = \text{mass} / \text{volume}$$

Binarization and Discretization

- Some data mining algorithms require that the data be in the form of binary attributes or categorical attributes

Binarization

Transforming continuous and discrete attributes into binary attributes

Simple Technique:

- If there are m categorical values, then uniquely assign each original value to an integer in the interval $[0, m - 1]$
- If the attribute is ordinal, then order must be maintained by the assignment
- Convert each of these m integers to a binary number
- Example: categorical variable with 5 values {awful, poor, OK, good, great} would require three binary variables x_1, x_2, x_3

Table 2.5. Conversion of a categorical attribute to three binary attributes.

Categorical Value	Integer Value	x_1	x_2	x_3
<i>awful</i>	0	0	0	0
<i>poor</i>	1	0	0	1
<i>OK</i>	2	0	1	0
<i>good</i>	3	0	1	1
<i>great</i>	4	1	0	0

Table 2.6. Conversion of a categorical attribute to five asymmetric binary attributes.

Categorical Value	Integer Value	x_1	x_2	x_3	x_4	x_5
<i>awful</i>	0	1	0	0	0	0
<i>poor</i>	1	0	1	0	0	0
<i>OK</i>	2	0	0	1	0	0
<i>good</i>	3	0	0	0	1	0
<i>great</i>	4	0	0	0	0	1

- Creates unintended relationships among the transformed attributes
attributes **x2 and x3 are correlated** because information about the **good** value is encoded using both attributes
- Association analysis requires asymmetric binary attributes, where only the presence of the attribute (value : 1) is important.
- For association problems, it is therefore necessary to introduce one binary attribute for each categorical value

Discretization

- Transformation of a continuous attribute to a categorical attribute

Involves two subtasks:

- Deciding how many categories to have:

After the values of the continuous attribute are sorted, they are then divided into n intervals by specifying $n - 1$ split points

- Determining how to map the values of the continuous attribute to these categories:

All the values in one interval are mapped to the same categorical value

Problem of discretization is one of deciding how many split points to choose and where to place them

Unsupervised Discretization

- Class information is not used

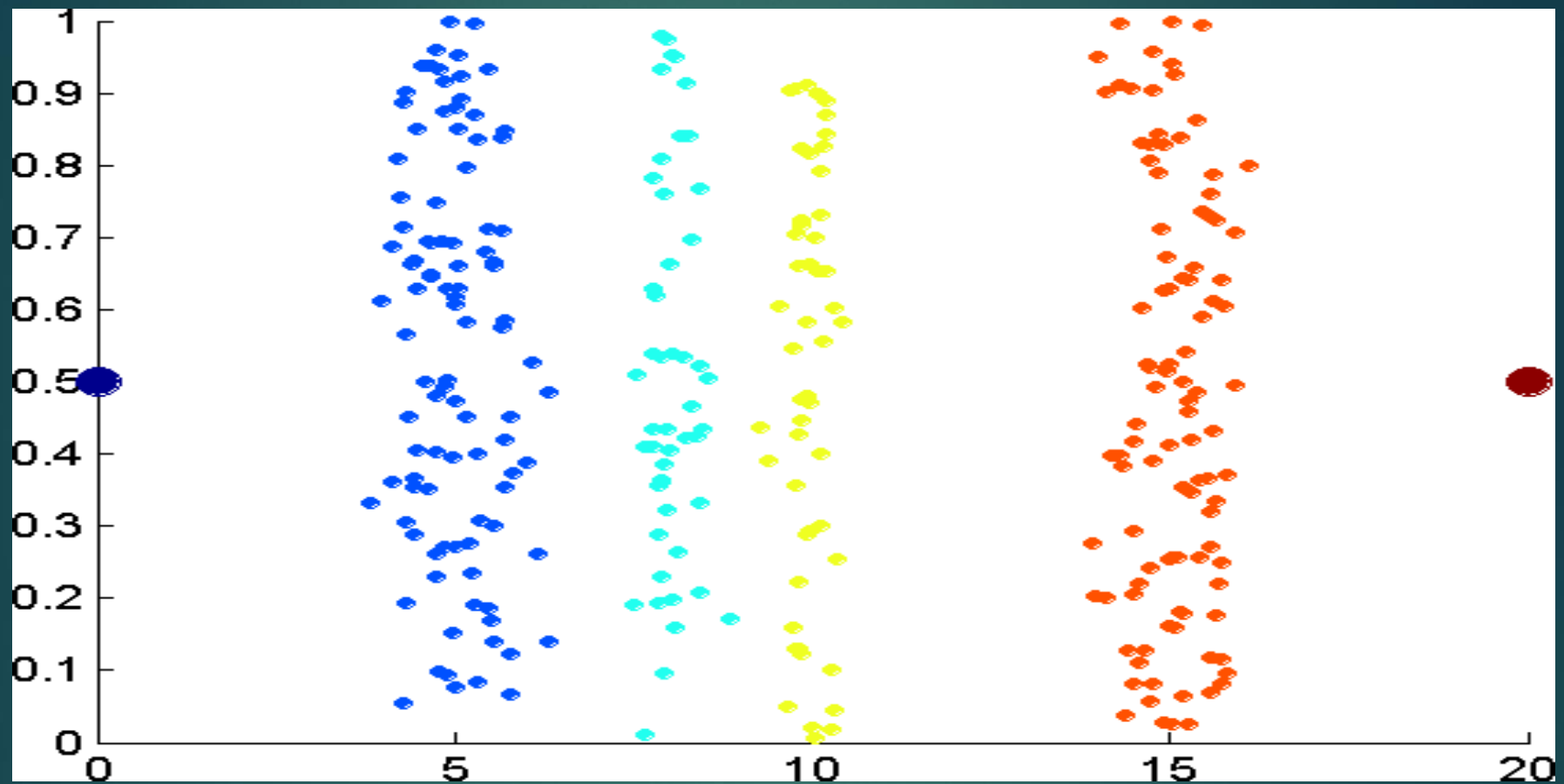
Equal width approach:

- divides the range of the attribute into a user-specified number of intervals each having the same width
- Can be badly affected by outliers

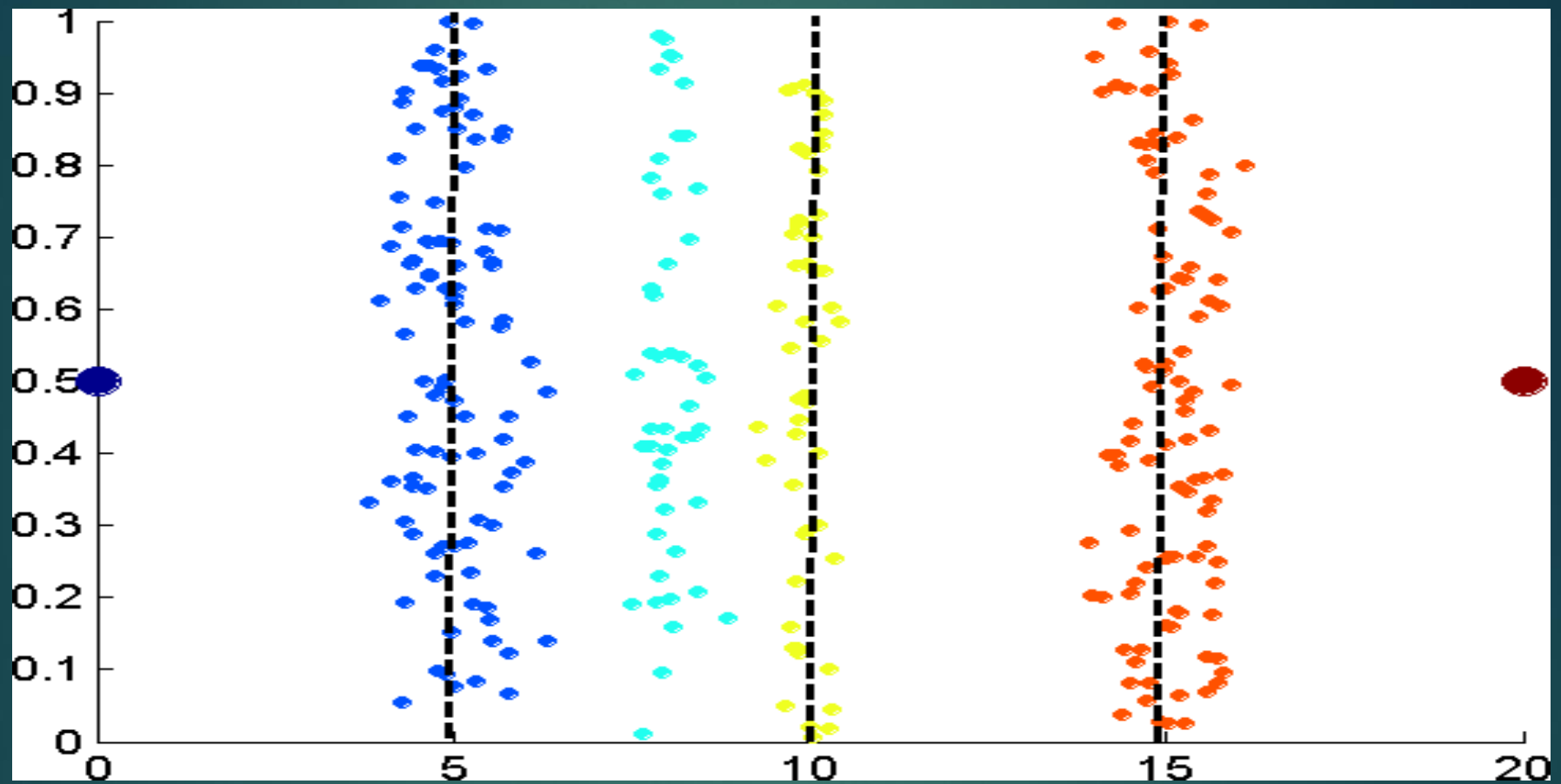


Equal frequency (equal depth) approach

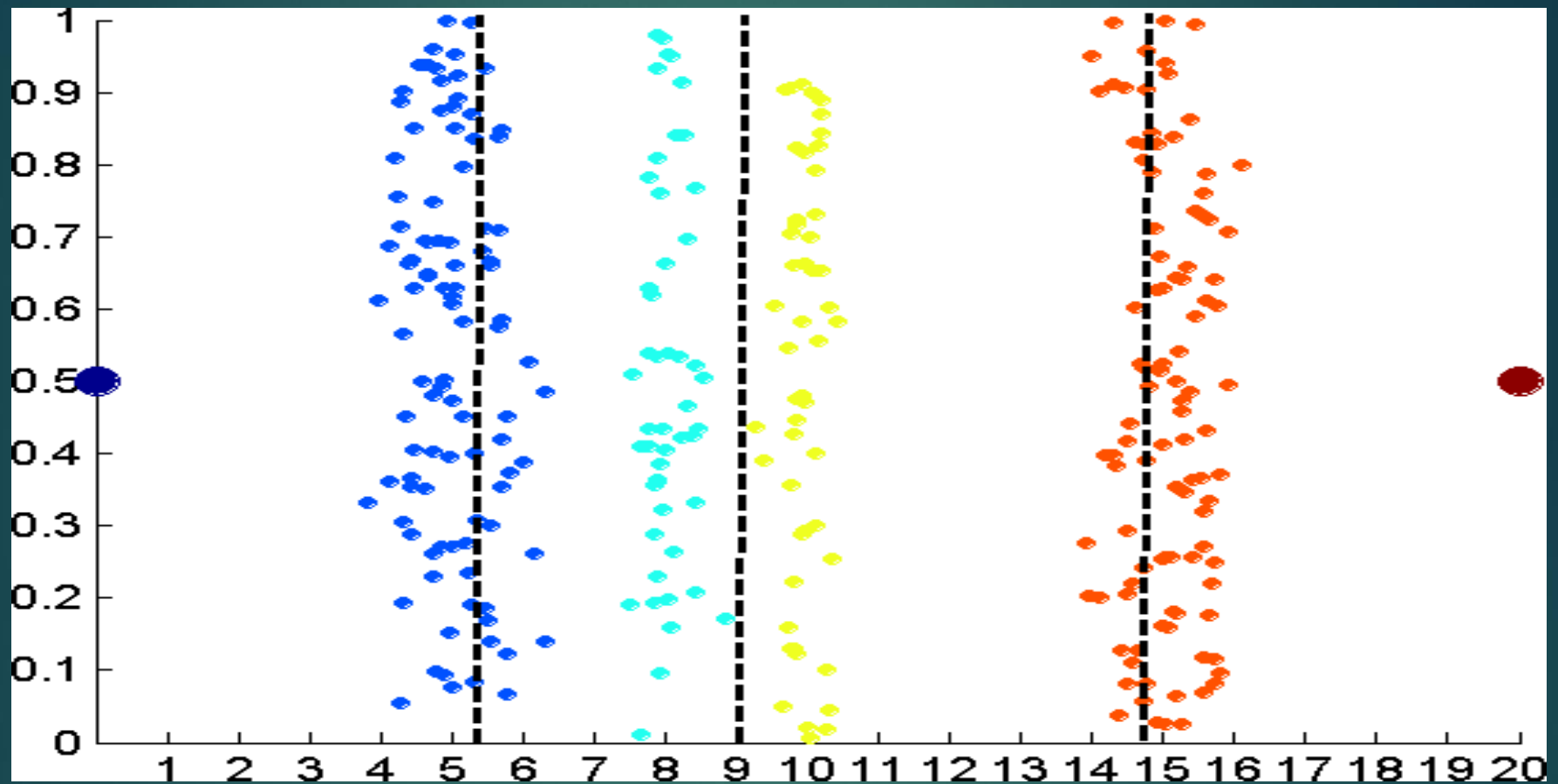
- Tries to put the same number of objects into each interval



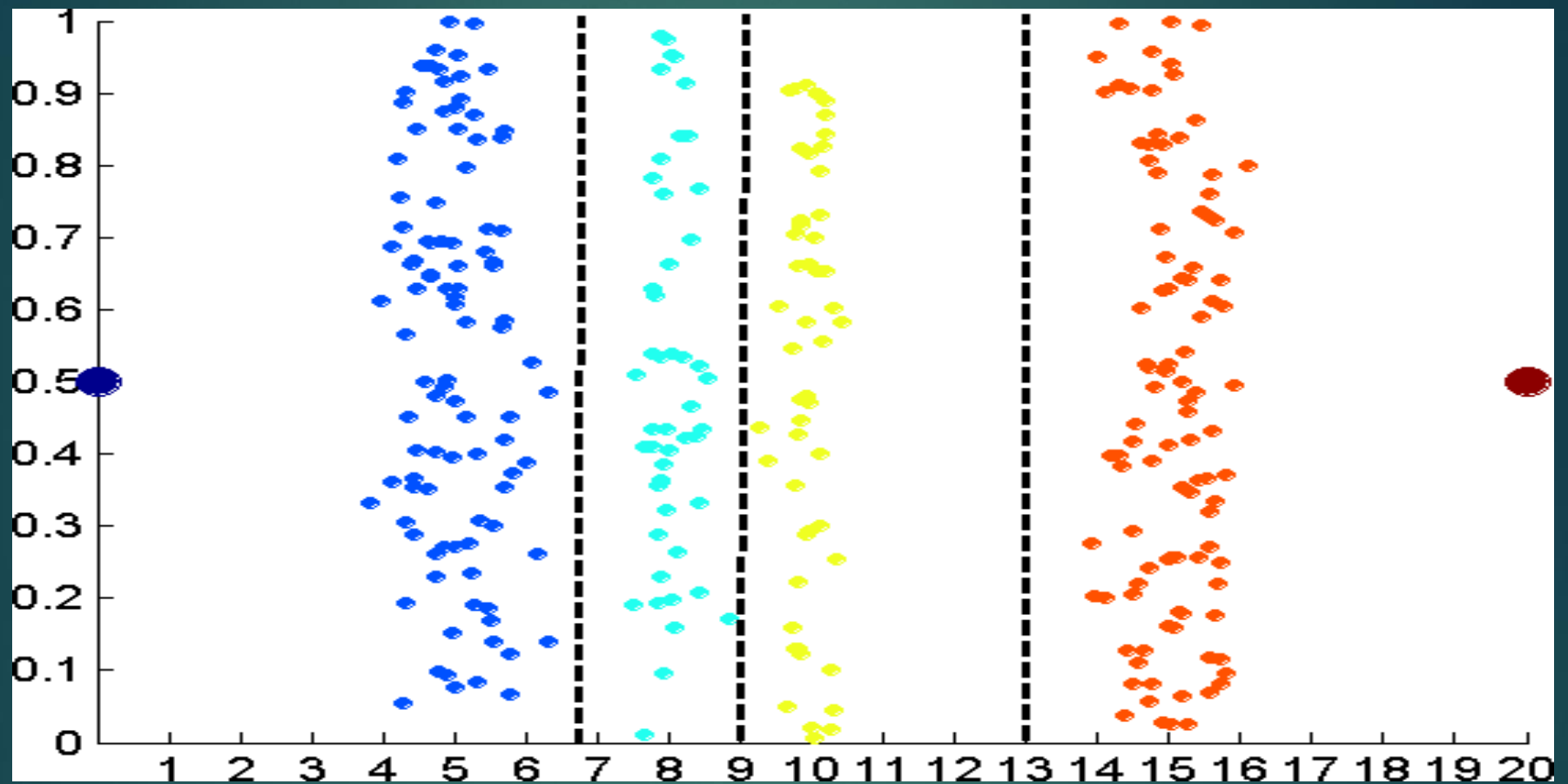
Data consists of four groups of points and two outliers. Data is one-dimensional, but a random y component is added to reduce overlap.



Equal interval width approach used to obtain 4 values.



Equal frequency approach used to obtain 4 values.



K-means approach to obtain 4 values

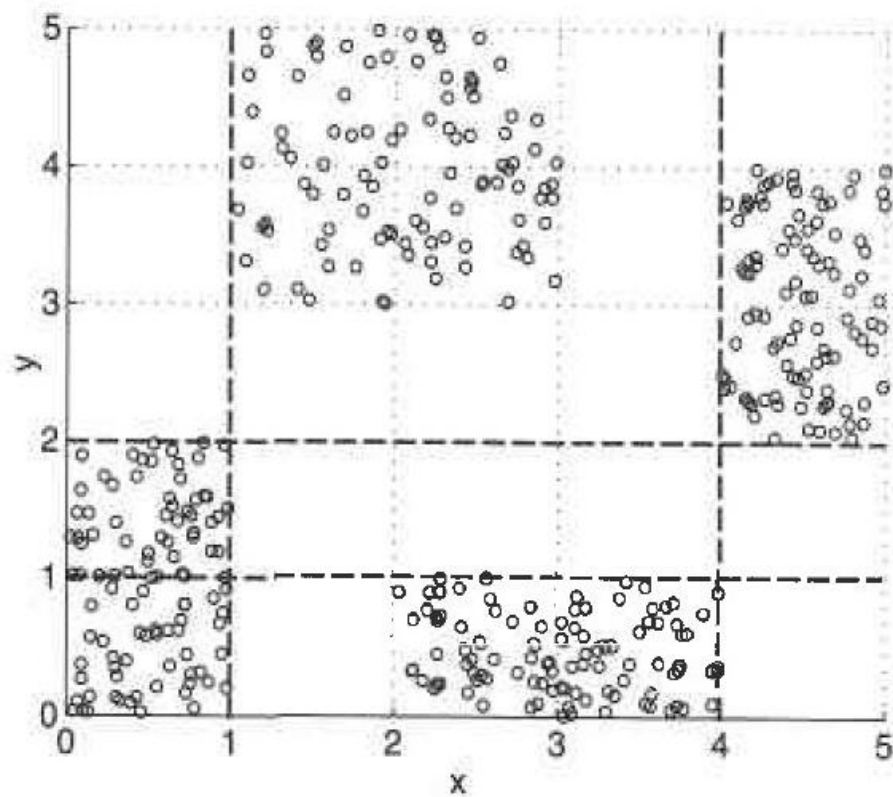
Supervised Discretization

- Class information is used
- An interval constructed with no knowledge of class labels often contains a mixture of class labels
- A conceptually simple approach is to place the splits in a way that maximizes the purity of the intervals
- Entropy based approaches are one of the most promising approaches to discretization

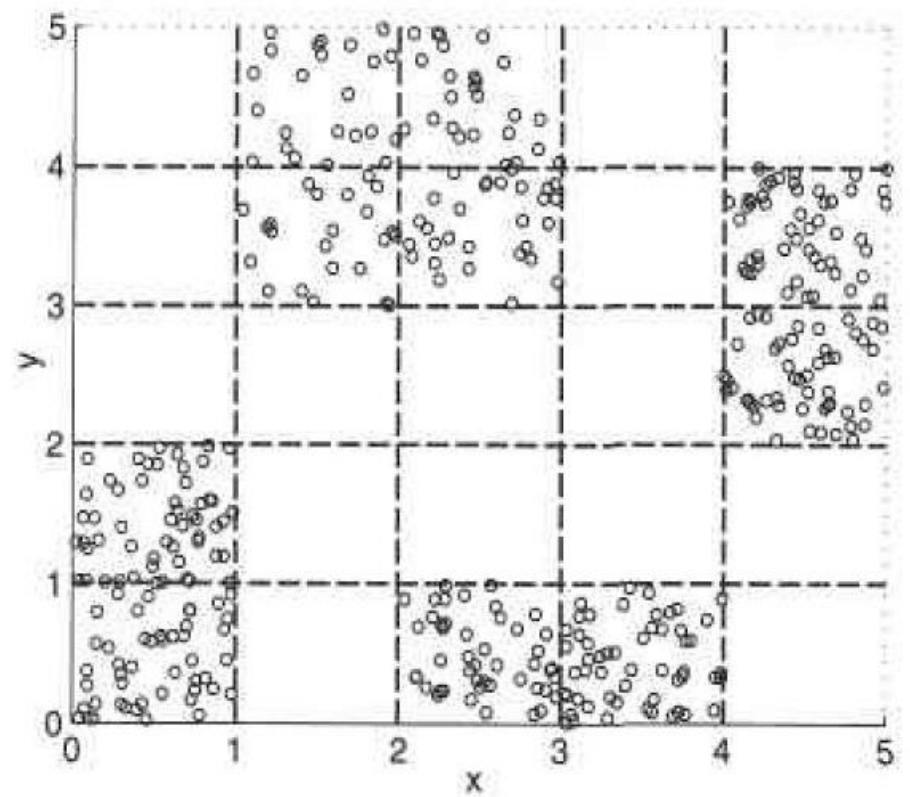
- Entropy of an interval is a measure of the purity of an interval
- If an interval contains only values of one class (is perfectly pure), then the entropy is 0
- If the classes of values in an interval occur equally often (the interval is as impure as possible), then the entropy is a maximum

Approach for partitioning a continuous attribute

- Bisecting the initial values so that the resulting two intervals give minimum entropy
- This technique only needs to consider each value as a possible split point, because it is assumed that intervals contain ordered sets of values
- Splitting process is then repeated with another interval, typically choosing the interval with the worst (highest) entropy, until a user-specified number of intervals is reached, or a stopping criterion is satisfied



(a) Three intervals



(b) Five intervals

Variable Transformation

- Transformation that is applied to all the values of a variable
- Variable transformations should be applied with caution since they change the nature of the data

Simple Functions

- Simple mathematical function is applied to each value individually
- If x is a variable, then examples of such transformations include x^k , $\log x$, e^x , square root of x , $1/x$, $\sin x$, or $|x|$ etc.

Normalization or Standardization

- Goal of standardization or normalization is to make an entire set of values have a particular property
- Min-max normalization: Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000, respectively.
- We would like to map *income* to the range [0.0, 1.0]. By min-max normalization, a value of \$73,600 for *income* is

$$\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716.$$